

Wei (David) Dai

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Education

Virginia Tech (Virginia Polytechnic Institute and State University)

Blacksburg, VA

Bachelor of Science in Computer Engineering

Aug. 2025 – May 2028

Coursework: Machine Learning, Principles of Computer Architecture, Applied Software Design, Signals and Systems

Purdue University Fort Wayne

Fort Wayne, IN

Bachelor of Science in Computer Engineering. GPA: 3.65/4.0

Aug. 2023 – May 2025

Honor Student & Dean's List & Honors List

Skills

Technical: C++, C, Python, Java, JavaScript/TypeScript, SQL, Qt, Git, Docker, CMake, PostgreSQL, Arduino

Language: Chinese - first language, English - Toefl: 101 (2021.08), Spanish – Intermediate level

Internships & Jobs

COSCO SHIPPING TECHNOLOGY Co., Ltd

Shanghai, China

Artificial Intelligence Application Division Intern

Jun. 2026 – Aug. 2026

- Designed a marine dual-spectrum PTZ surveillance solution integrating visible-light, thermal imaging, AIS, GNSS/RTK, and AI perception.
- Evaluated hardware, optical, and embedded AI configurations and finalized the system requirements and product design.
- Delivered the final camera solution for manufacturing; the prototype is currently in production.

Purdue University Fort Wayne

Fort Wayne, IN

Student IT Support Specialist

Nov. 2024 – May 2025

- Delivered front-line technical support to 15,000+ students, faculty, and staff, resolving hardware, software, account, and campus system issues.
- Diagnosed and troubleshooted Windows, macOS, and network connectivity problems.

Shanghai Automotive Industry Corporation – Utopilot

Shanghai, China

Perception Department Intern

Jun. 2024 – Aug. 2024

- Analyzed autonomous-driving perception systems for heavy-duty trucks in port and mining environments, focusing on obstacle detection and environmental mapping.
- Investigated point-cloud processing and occupancy-grid (OCC) methods for representing camera and sensor data in 3D spatial environments.

Projects

AI Workspace – C++ Desktop Retrieval & Processing System

Aug. 2026 – Present

- Developing a modular C++ desktop application for document ingestion, chunking, corpus indexing, TF-IDF retrieval, and context construction, with an extensible architecture using polymorphism and strategy patterns
- Building an event-driven Qt interface with separate processing services using inter-process communication (IPC), asynchronous tasks, CMake, Docker, and automated testing for responsive background processing

VDM Ledger — Full-Stack Personal Finance & Banking Application

Sep. 2026 – Present

- Built and deployed a full-stack finance application with Next.js, TypeScript, Supabase/PostgreSQL, and Plaid, integrating bank accounts, transaction syncing, financial dashboards, and Excel reporting; hosted on Vercel.
- Designed layered architecture for UI, business services, and database repositories; implemented authenticated APIs, PostgreSQL row-level security, scheduled jobs, and GitHub Actions CI with automated testing and production build validation.

Personal Smart Locker – Embedded Access Control System

Aug. 2026 – Present

- Built an embedded smart locker using an ATmega328P, ADC keypad, servo, LEDs, and buzzer, with state-machine-based PIN entry, lock control, and sleep/wake behavior
- Developed C/C++ firmware for keypad calibration, servo PWM, EEPROM password storage, and stall recovery; validated operation through power-cycle and system testing

Leadership & Activities

IEEEExtreme 17.0 Competition

Team Leader | Nov. 2023

- Led a team to 1st place at Purdue and 51st in the U.S. in a 24-hour global programming competition.